Pimpri Chinchwad Education Trust's

Pimpri Chinchwad College of Engineering

Department of Information Technology

ASSIGNMENT 7

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MongoDB – CRUD Operations & Joins

1. CREATE (SAVE / INSERT)

Insert into departments

```
db.departments.insertMany([
  { dept_id: 1, dept_name: "HR" },
  { dept_id: 2, dept_name: "Engineering" }
])
```

Insert into employees

```
db.employees.insertMany([
  { emp_id: 101, name: "Amit", age: 30, dept_id: 1 },
  { emp_id: 102, name: "Neha", age: 28, dept_id: 2 }
])
```

Insert into appraisals

```
db.appraisals.insertMany([
  { appraisal_id: 201, emp_id: 101, rating: 4.5, year: 2024 },
  { appraisal_id: 202, emp_id: 102, rating: 3.8, year: 2024 }
])
```

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2. READ (SELECT with Logical Operators)

Basic Find

```
db.employees.find()
```

Using AND

```
db.employees.find({
  age: { $gt: 28 },
  dept_id: 2
})
```

Using OR

```
db.employees.find({
  $or: [
    { name: "Amit" },
    { age: { $lt: 30 } }
  ]
})
```

Using NOT

```
db.employees.find({
  dept_id: { $ne: 1 }
})
```

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3. UPDATE

Update One

```
db.employees.updateOne(
  { emp_id: 101 },
  { $set: { age: 31 } }
)
```

Update Many

```
db.employees.updateMany(
  { dept_id: 2 },
  { $set: { age: 29 } }
)
```

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4. DELETE

Delete One

```
db.employees.deleteOne({ emp_id: 102 })
```

Delete Many

```
db.employees.deleteMany({ dept_id: 2 })
```

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5. JOINS in MongoDB (Using $lookup)

Join Employees with Departments

```
db.employees.aggregate([
  {
```

```
    $lookup: {
      from: "departments",
      localField: "dept_id",
      foreignField: "dept_id",
      as: "department_info"
    }
  }
])
```

Join Multiple Collections (Employees + Departments + Appraisals)

```
db.employees.aggregate([
  {
    $lookup: {
      from: "departments",
      localField: "dept_id",
      foreignField: "dept_id",
      as: "dept"
    }
  },
  {
    $lookup: {
      from: "appraisals",
      localField: "emp_id",
      foreignField: "emp_id",
      as: "appraisals"
    }
  }
])
```

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6. Additional (SAVE Method / UPSERT)

Insert or Update (Save Equivalent)

```
db.employees.updateOne(
  { emp_id: 103 },
  {
    $set: {
      name: "Rahul",
      age: 25,
      dept_id: 1
    }
  },
  { upsert: true }
)
```

Summary

Operation	MongoDB Method
CREATE	insertOne(), insertMany()
READ	find()
UPDATE	updateOne(), updateMany()
DELETE	deleteOne(), deleteMany()
JOIN	$lookup
SAVE	upsert: true